EE 798I: NANOPHOTONICS **Assignment 4** **2025-26 First Semester**

Q.1) **Lossy waveguide** **(50 marks)**

Consider the fundamental TE mode of a 400 nm thick dielectric slab waveguide made of Silicon at 800 nm wavelength (free space). You can find out the complex refractive index of Si on the link given below. Determine and plot the electric and magnetic field phasors throughout the structure. Also determine and plot the Poynting vector components. What is the propagation length? Compare the results with the case of lossless medium as discussed in class.

Link: https://refractiveindex.info/?shelf=main&book=Si&page=Aspnes

Q.2) **TM modes of a dielectric slab waveguide** **(50 marks)**

Revisit the derivation TE even and odd modes of a dielectric slab of finite thickness (surrounded by a lower index medium on both sides). Repeat the same for TM polarized guided modes. Derive the dispersion relations and compare them with those of the TE polarized modes. Write the expressions for all the non-zero field components (Hx, Ey, Ez).

Q.2) **Guided modes and their sources** **(50 marks)**

Consider the fundamental TM mode of a lossless dielectric slab waveguide. There is an oscillating surface current $J_S(z)\,\hat{x}$ in the y = 0 plane which excites outward propagating fundamental TM mode on both sides with equal amplitude. Determine the current distribution as a function of the excited TM mode amplitude. Does the current source emit or absorb net power? If it does, where is the energy coming from or vanishing to?

Hint: Work in the frequency domain (phasors) and employ the field boundary conditions to figure out a solution.

Q.3) **Diffraction grating** **(50 marks)**

Design and verify using COMSOL, a 1D diffraction grating made of Silicon which retro-reflects a plane wave incident at $\pi/4$, i.e., there should be a diffracted plane wave at $-\pi/4$.

BONUS QUESTION: **Guided modes of a waveguide with finite cross-section** **(100 marks)**

Consider a Silicon waveguide of finite cross-section ($W \times H$) with a high aspect ratio ($W \gg H$). Find out the modes of this waveguide by using mode analysis method in COMSOL. You should choose the dimensions of the waveguide to ensure single mode operation at free space wavelength of 1550 nm. Compare the numerically obtained field distribution with that of the dielectric slab waveguide obtained analytically.